THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/1 **BIOLOGY 1**

(For Both School and Private Candidates)

Duration: 3 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** the questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. All writing should be in **blue** or **black** pen, **except** drawing which must be in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet(s).

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SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) - (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

(i) Farmers at Mtakuja village observed a continuous decrease of amount of beans yield in their farms. How is this step in the scientific process refers?

A Experimentation B Problem identification C Hypothesis
D Conclusion E Data collection

(ii) Form Two students who recently visited Mikumi National Park saw Zebra, Shrubs, Giraffe, Lion, Cheetah, Grasses, and Mushroom. Which of these organisms would they group as primary consumers?

A Grasses and shrubs B Giraffe and shrubs C Lion and cheetah
D Zebra and giraffe E Mushroom and zebra

(iii) In the process of making doughnuts a woman mixed wheat flour, yeasts, sugar, and water; kneaded the mixture and covered it. After half an hour, the dough raised. Which one of the following made the dough to rise?

A Yeasts B Glucose C Wheat flour
D Water E Sugar

(iv) A bleeding pregnant woman was confirmed to have an embryo implanted in the fallopian tube. Which complication was she suffering from?

A Preeclampsia B Miscarriage
C Breech birth D Premature birth
E Ectopic pregnancy

(v) Which mechanism does the human body undergo in response to cold weather condition?

A Vasodilation of arterioles B Vasoconstriction of arterioles
C Sweating and panting D Relaxation of erector muscles
E Decreased metabolic rate

(vi) Which diseases occur worldwide?

A Tuberculosis and cholera B Malaria and bilharzia
C COVID - 19 and cholera D AIDS and COVID - 19
E AIDS and tuberculosis

(vii) If you have noticed water spills on the floor of the class room, what would you do to prevent accident?

A Wait for the floor to dry
B Walk slowly on the floor
C Wipe the floor immediately
D Apply mud on the floor
E Shift to another classroom

(viii) Which one of the following is **not** a part of a movable joint?

A Bone
B Tendon
C Cartilage
D Synovial fluid
E Ligament

(ix) Which one of the following are examples of sex linked characteristics in human?

A Colour blindness and albinism
B Haemophilia and tongue rolling
C Colour blindness and haemophilia
D Tongue rolling and blood group
E Blood group and haemophilia

(x) Organisms in class Reptilia have dry skin with horny scales. Which of the following organisms fall under this category?

A Lizard, crocodile, turtle and snake
B Crocodile, bat, chicken and lizard
C Snake, salamander, frog and crocodile
D Turtle, mouse, snake and salamander
E Eagle, flamingo, chicken and lizard

2. Match the roles of the hormones in **List A** with their corresponding hormones in **List B** by writing the letter of the correct response beside the item number in the answer booklet(s) provided.

List A		**List B**	
(i)	Controls contraction and relaxation of uterine walls during birth.	A	Antidiuretic
		B	Thyroxine
(ii)	Promotes conversion of excess glucose into glycogen in the blood.	C	Adrenaline
		D	Oxytocin
(iii)	Regulates amount of water reabsorption in the kidney tubules.	E	Glucagon
		F	Insulin
(iv)	Stimulates milk production in lactating mammals.	G	Oestrogen
(v)	Controls metabolic activities in the body.	H	Prolactin
(vi)	Promotes development of secondary sexual characteristics in females.		

SECTION B (54 Marks)

Answer **all** questions in this section.

3. Mkutano villagers use traditional methods to process, preserve and store variety of foods they obtain from their daily activities. Outline six advantages of such methods.

4. With the aid of a labeled diagram, describe the events which take place during the prophase stage of mitosis.

5. How are the cardiac muscles adapted to perform their roles? Give six points.

6. Bacteria, Amoeba and Non-flowering plants reproduce asexually. Explain three merits and three demerits of the type of reproduction exhibited by these organisms.

7. A man with heterozygous blood group A married a woman with heterozygous blood group B. One among their offsprings produced was blood group O. Use genetic cross to show the probability of producing such offspring.

8. By using example, explain how each of the following provide evidences for organic evolution. Give three points for each.
   (a) Fossil records
   (b) Comparative embryology

SECTION C (30 Marks)

Answer **two (2)** questions from this section.

9. You are invited as a guest speaker in a youth meeting to fight against drug abuse. Recommend six preventive and control measures you would address to them.

10. Matatu villagers have appointed you as a chair person for campaign against gonorrhoea prevention. Explain three ways of transmission and five measures that will help them to prevent the disease.

11. Analyse five environmental factors which affect the rate of transpiration in plants.

THE UNITED REPUBLIC OF TANZANIA
THE NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/2A

BIOLOGY 2A
(ACTUAL PRACTICAL A)
(For Both School and Private Candidates)

Duration: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writings should be in **blue** or **black** ink, except diagrams which must be in pencil.
4. Communication devises and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).

2 501033 221021

1. You have been provided with specimens **O** and **M**, the chemical reagents and apparati. You are required to identify the food substances present in the specimens.

(a) Prepare the solutions from specimen **O** and **M** for the food test experiments then, outline the procedures used to prepare each solution.

(b) Carry out experiments to identify the food substances present in specimens **O** and **M**. Your experimental report should be tabulated as shown in the following Table:

Identification of Food Substances in Specimens **O** and **M**

Food Tested	**Procedure**	**Observation**	**Inference**

(c) State the function(s) of each food substance identified in 1(b) to the human body.

(d) Apart from the specimen provided, give another one example of natural source from which the food substances identified in 1(b) could be obtained.

2. You have been provided with specimens **E**, **F** and **G** obtained from different habitat. Carefully study them and answer the following questions:

(a) Identify the Phylum/Division and the Kingdom in which each specimen **E**, **F** and **G** belongs.

(b) For each of specimens **E**, **F** and **G** state three observable features which are characteristic of its respective Phylum/Division.

(c) Name the habitat of each of specimens **E**, **F** and **G**.

(d) For each of specimens **E**, **F** and **G** state whether it is a producer, consumer or decomposer in its natural habitat. Give reason for each case.

(e) State two advantages and one disadvantage of specimen **E** to other living organisms.

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033/2B

BIOLOGY 2B
(ACTUAL PRACTICAL B)
(For Both School and Private Candidates)

Duration: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writings should be in **blue** or **black** ink, except diagrams which must be in pencil.
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5. Write your **Examination Number** on every page of your answer booklet(s).

2 501033 222028

1. (a) You have been provided with solutions **P** and **Q** each containing a mixture of food substances. Carry out the experiment to identify the food substances present in the solutions provided. Your experimental report should be tabulated as shown in the Table 1.

Table 1: Identification of Food Substances in the Solutions **P** and **Q**

Food Tested	Procedure	Observation	Inference

(b) Specimen **N** contains lipid among others. Prepare the specimen for the lipid test experiment using a filter paper then, outline the procedure used to prepare it.

(c) Use the filter paper provided to identify the presence of lipid in the specimen **N**. Present your findings as shown in Table 2.

Table 2: Identification of Lipid in Specimen **N**

Food Tested	Procedure	Observation	Inference

(d) State the function(s) of each food substance identified in the solutions and specimen **N** to the human body.

(e) Why boiling was important in some procedures of the experiments conducted?

2. You have been provided with specimens **I**, **J**, **K** and **L**. Carefully, study them and answer the following questions:

(a) State two distinctive features of each specimen **I**, **J** and **K** at Class level.

(b) Specimens **I** and **J** belong to the same major group. State two characteristics they have in common.

(c) How do the specimens **I** and **L** adopted to survive in their natural habitats? Give one point for each.

(d) State two ways the farmers benefit from the availability of the specimen **L** in their farms.

(e) Outline three advantages and one disadvantage of the specimen **I** in nature.

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033/2C

BIOLOGY 2C
(ACTUAL PRACTICAL C)
(For Both School and Private Candidates)

Duration: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writings should be in **blue** or **black** ink, except diagrams which must be in pencil.
4. Communication devises and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).

1. (a) You have been provided with solutions **A** and **B** which contain the mixture of different food substances. Carry out an experiment to identify the food substances present in each solution. Your experimental report should be tabulated as shown in Table 1.

Table 1: Identification of Food Substances in Solution **A** and **B**

Food tested	Procedure	Observation	Inference

(b) Specimen **C** contains protein among others. Prepare the solution from it for Biuret test experiment then, state the procedure followed to prepare it.

(c) Carry out the Biuret test to identify food substances present in specimen **C**. Your experimental report should be tabulated as shown in Table 2.

Table 2: Identification of Biuret Test for Food in Specimen **C**

Food Tested	Procedure	Observation	Inference

(d) State the function(s) of each food substances identified in solutions **A**, **B** and specimen **C** to the human body.

(e) "Sodium hydroxide" and "dilute hydrochloric acid" were important in some procedure of the experiments. Justify this statement.

2. You have been provided with specimens **W**, **X**, **Y** and **Z** obtained from different habitats. Carefully study them then, answer the following questions:

(a) Identify the Kingdom, Phylum/Division and Class in which each specimen **W**, **X**, **Y** and **Z** belongs.

(b) For each specimen **W**, **X**, **Y** and **Z**, state three visible futures which are the characteristics of the respective Class it belongs.

(c) Specimens **W** and **X** belongs to the same Kingdom and Phylum. Which characteristics they have in common to represent those groups?

(d) For each specimen **W** and **Y** state two advantages and one disadvantage they have for other living organisms.

THE UNITED REPUBLIC OF TANZANIA

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CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/1 **BIOLOGY 1**

(For Both School and Private Candidates)

Duration: 3 Hours **SOLUTIONS** **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C **thirty (30)** marks.
4. All writing must be in **blue** or **black** pen, except drawing which must be in pencil.
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SECTION A

1. For each of the items (i) to (x), the question is copied, alternatives are copied, then the correct answer is given with a reason.

(i) Farmers at Mtakuja village observed a continuous decrease of amount of beans yield in their farms. How is this step in the scientific process refers?

A Experimentation
B Problem identification
C Hypothesis
D Conclusion
E Data collection

Correct answer: B Problem identification.
Reason: Observing a continuous decrease in yield is the recognition of a problem that needs investigation, which is the problem identification stage of the scientific process.

(ii) Form Two students who recently visited Mikumi National Park saw Zebra Shrubs Giraffe Lion Cheetah Grasses and Mushroom. Which of these organisms would they group as primary consumers?

A Grasses and shrubs
B Giraffe and shrubs
C Lion and cheetah
D Zebra and giraffe
E Mushroom and zebra

Correct answer: D Zebra and giraffe.
Reason: Primary consumers are herbivores that feed directly on producers. Zebra and giraffe feed on grasses and shrubs.

(iii) In the process of making doughnuts a woman mixed wheat flour yeasts sugar and water kneaded the mixture and covered it. After half an hour the dough raised. Which one of the following made the dough to rise?

A Yeasts
B Glucose
C Wheat flour
D Water
E Sugar

Correct answer: A Yeasts.
Reason: Yeast ferments sugars and produces carbon dioxide gas, which causes the dough to rise.

(iv) A bleeding pregnant woman was confirmed to have an embryo implanted in the fallopian tube. Which complication was she suffering from?

A Preeclampsia
B Miscarriage
C Breech birth
D Premature birth
E Ectopic pregnancy

Correct answer: E Ectopic pregnancy.
Reason: Implantation of an embryo outside the uterus, especially in the fallopian tube, is known as ectopic pregnancy.

(v) Which mechanism does the human body undergo in response to cold weather condition?

A Vasodilation of arterioles
B Vasoconstriction of arterioles

C Sweating and panting
D Relaxation of erector muscles
E Decreased metabolic rate

Correct answer: B Vasoconstriction of arterioles.
Reason: Vasoconstriction reduces blood flow to the skin, minimizing heat loss in cold conditions.

(vi) Which diseases occur worldwide?

A Tuberculosis and cholera
B Malaria and bilharzia
C COVID 19 and cholera
D AIDS and COVID 19
E AIDS and tuberculosis

Correct answer: D AIDS and COVID 19.
Reason: Pandemic diseases are those that occur worldwide, and both AIDS and COVID 19 affect populations globally.

(vii) If you have noticed water spills on the floor of the class room what would you do to prevent accident?

A Wait for the floor to dry
B Walk slowly on the floor
C Wipe the floor immediately
D Apply mud on the floor
E Shift to another classroom

Correct answer: C Wipe the floor immediately.
Reason: Removing the water immediately eliminates the risk of slipping and prevents accidents.

(viii) Which one of the following is not a part of a movable joint?

A Bone

B Tendon

C Cartilage

D Synovial fluid

E Ligament

Correct answer: B Tendon.

Reason: Tendons connect muscles to bones and are not structural components of movable joints.

(ix) Which one of the following are examples of sex linked characteristics in human?

A Colour blindness and albinism

B Haemophilia and tongue rolling

C Colour blindness and haemophilia

D Tongue rolling and blood group

E Blood group and haemophilia

Correct answer: C Colour blindness and haemophilia.

Reason: Both traits are controlled by genes located on the X chromosome.

(x) Organisms in class Reptilia have dry skin with horny scales. Which of the following organisms fall under this category?

A Lizard crocodile turtle and snake

B Crocodile bat chicken and lizard

C Snake salamander frog and crocodile

D Turtle mouse snake and salamander

E Eagle flamingo chicken and lizard

Correct answer: A Lizard crocodile turtle and snake.

Reason: These organisms are reptiles characterized by dry skin covered with horny scales.

2. Match the roles of the hormones in **List A** with their corresponding hormones in **List B** by writing the letter of the correct response beside the item number in the answer booklet(s) provided.

List A	**List B**
(i) Controls contraction and relaxation of uterine walls during birth.	A Antidiuretic
(ii) Promotes conversion of excess glucose into glycogen in the blood.	B Thyroxine
(iii) Regulates amount of water reabsorption in the kidney tubules.	C Adrenaline
(iv) Stimulates milk production in lactating mammals.	D Oxytocin
(v) Controls metabolic activities in the body.	E Glucagon
(vi) Promotes development of secondary sexual characteristics in females.	F Insulin
	G Oestrogen
	H Prolactin

Answers

(i)	**(ii)**	**(iii)**	**(iv)**	**(v)**	**(vi)**
D	F	A	H	B	G

SECTION B

3. Mkutano villagers use traditional methods to process preserve and store variety of foods they obtain from their daily activities. Outline six advantages of such methods.

Traditional food preservation methods are cheap and affordable because they rely on locally available materials and simple techniques.

They help extend the shelf life of foods by reducing spoilage and wastage.

Traditional methods preserve food flavor and nutritional value when properly applied.

They do not require electricity, making them suitable for rural areas.

These methods promote food security during periods of scarcity.

They preserve cultural knowledge and indigenous practices passed through generations.

4. With the aid of a labeled diagram describe the events which take place during the prophase stage of mitosis.

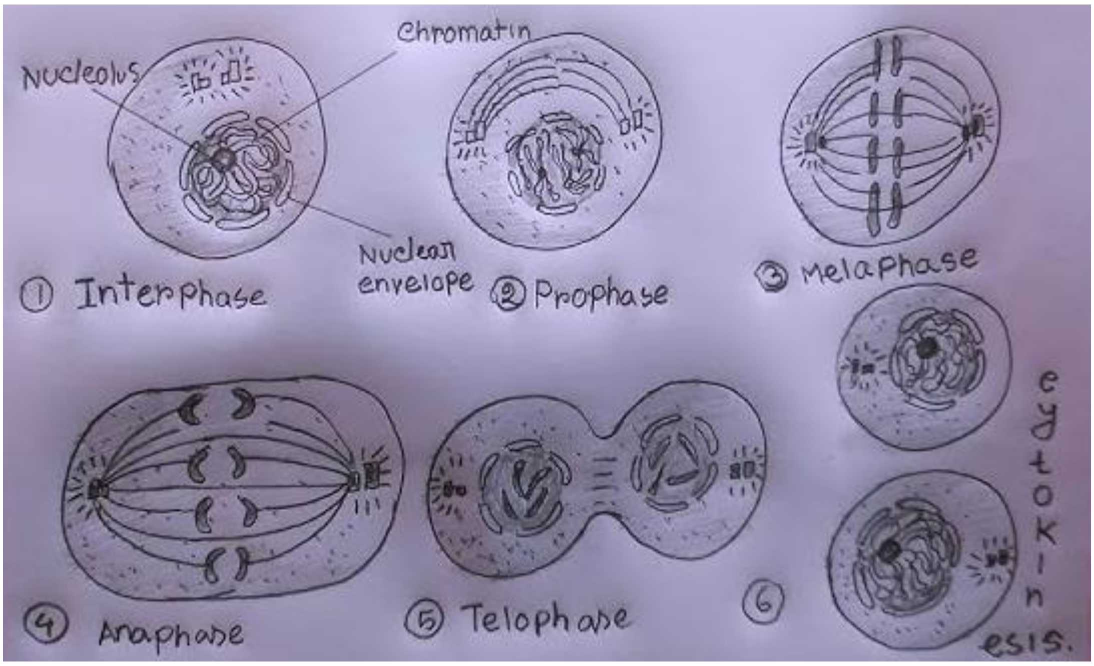
During prophase, chromatin fibers condense to form visible chromosomes, each consisting of two sister chromatids joined at the centromere.

The nuclear membrane gradually breaks down, allowing spindle fibers to interact with chromosomes.

The nucleolus disappears, indicating the cell is preparing for division.

Spindle fibers begin to form from centrioles and extend toward opposite poles of the cell.

Chromosomes start moving toward the equatorial plane under the influence of spindle fibers.



5. How are the cardiac muscles adapted to perform their roles? Give six points.

Cardiac muscles are branched, allowing strong and coordinated contractions.

They are interconnected by intercalated discs, which ensure rapid transmission of impulses.

They are rich in mitochondria, providing sufficient energy for continuous activity.

Cardiac muscles are involuntary, allowing the heart to function without conscious control.

They have a rich blood supply to provide oxygen and nutrients.

They are fatigue resistant, enabling the heart to beat continuously throughout life.

6. Bacteria Amoeba and Non flowering plants reproduce asexually. Explain three merits and three demerits of the type of reproduction exhibited by these organisms.

Asexual reproduction is fast, allowing rapid population increase.

It requires only one parent, making reproduction possible in isolation.

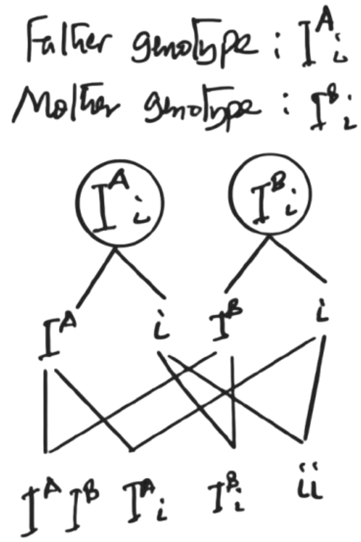
It conserves energy since mating is not required.

However, it produces no genetic variation, making organisms vulnerable to diseases.

Unfavorable conditions can wipe out entire populations due to identical traits.

It limits evolutionary adaptation to changing environments.

7. A man with heterozygous blood group A married a woman with heterozygous blood group B. One among their offsprings produced was blood group O. Use genetic cross to show the probability of producing such offspring.



Father genotype: IAi
Mother genotype: IBi

Gametes from father: IA, i
Gametes from mother: IB, i

Offspring genotypes:
IAIB, IAi, IBi, ii

Probability of blood group O (ii) is 1 out of 4, which is 25 percent.

8. By using example explain how each of the following provide evidences for organic evolution. Give three points for each.

(a) Fossil records

Fossils show progressive changes in organisms over time, indicating gradual evolution.

They reveal extinct species that link past and present organisms.

Fossils help establish evolutionary timelines through geological layers.

(b) Comparative embryology

Early embryos of different species show similar structures, suggesting common ancestry.

Shared developmental stages indicate evolutionary relationships.

Differences appear later, showing divergence from a common origin.

SECTION C

9. You are invited as a guest speaker in a youth meeting to fight against drug abuse. Recommend six preventive and control measures you would address to them.

One important preventive measure against drug abuse is education and awareness. Youths should be taught about the health, social, and economic dangers of drug abuse so that they can make informed decisions and avoid experimenting with drugs.

Parental guidance and family support play a major role in preventing drug abuse. When parents communicate openly with their children, monitor their behavior, and provide emotional support, youths are less likely to engage in drug use.

Strict enforcement of laws controlling production, distribution, and use of drugs helps reduce availability. When laws are enforced effectively, access to drugs becomes limited, discouraging abuse among young people.

Provision of counseling and rehabilitation services is essential for control of drug abuse. Youths who are already addicted can receive psychological support and treatment to help them recover and reintegrate into society.

Engaging youths in sports, vocational training, and income generating activities helps reduce idleness. When young people are productively occupied, they are less likely to turn to drugs out of boredom or frustration.

Peer education and support groups also help prevent drug abuse. Positive peer influence encourages responsible behavior and helps youths resist pressure to use drugs.

10. Matatu villagers have appointed you as a chair person for campaign against gonorrhoea prevention. Explain three ways of transmission and five measures that will help them to prevent the disease.

Gonorrhoea is mainly transmitted through unprotected sexual intercourse with an infected person. The bacteria are passed from one individual to another through contact with infected body fluids during sexual activity.

The disease can also be transmitted from an infected mother to her baby during childbirth. This can result in serious health complications for the newborn, including eye infections and blindness.

Another mode of transmission is through sharing contaminated sex toys without proper cleaning. This allows the bacteria to move from one person to another.

One effective preventive measure is practicing safe sex through consistent and correct use of condoms. Condoms reduce direct contact with infected fluids, thereby lowering the risk of transmission.

Faithfulness to one uninfected sexual partner helps prevent gonorrhoea. Limiting the number of sexual partners reduces exposure to sexually transmitted infections.

Regular medical screening and testing enable early detection of infection. Early diagnosis allows timely treatment and prevents further spread of the disease.

Prompt treatment of infected individuals and their partners is essential. Treating all affected persons breaks the chain of transmission in the community.

Health education campaigns increase community awareness about gonorrhoea. When people understand symptoms, transmission, and prevention methods, they are more likely to protect themselves.

11. Analyse five environmental factors which affect the rate of transpiration in plants.

Temperature greatly influences the rate of transpiration. High temperatures increase evaporation of water from leaf surfaces, causing transpiration to occur at a faster rate.

Humidity affects transpiration by controlling the diffusion of water vapor. Low humidity increases transpiration because dry air allows more water vapor to move out of the leaf.

Wind speed also affects transpiration rate. Increased wind removes the moist air around leaves, maintaining a diffusion gradient that allows continuous water loss.

Light intensity influences transpiration by affecting stomatal opening. In bright light, stomata open wider to allow gas exchange, which increases water loss through transpiration.

Availability of water in the soil determines how much water a plant can absorb. When soil water is adequate, transpiration proceeds normally, but when water is scarce, stomata close and transpiration rate decreases.

THE UNITED REPUBLIC OF TANZANIA

NATIONAL EXAMINATIONS COUNCIL OF TANZANIA

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/2A **BIOLOGY 2A**

(ACTUAL PRACTICAL A)

(For Both School and Private Candidates)

Duration: 2:30 Hours **SOLUTIONS** **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writing should be in **blue** or **black** ink, except drawing which must be in pencil.
4. Communication devices and any unauthorised materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)

1. You have been provided with specimens O and M the chemical reagents and apparatus. You are required to identify the food substances present in the specimens.

(a) Prepare the solutions from specimen O and M for the food test experiments then outline the procedures used to prepare each solution.

Preparation of specimen O solution

A small amount of specimen O is crushed using a mortar and pestle. Distilled water is added and the mixture is stirred thoroughly. The mixture is then filtered to obtain a clear solution for food testing.

Preparation of specimen M solution

Specimen M is ground into small particles. Distilled water is added and the mixture is shaken well. The mixture is filtered to obtain a clear solution ready for food tests.

(b) Carry out experiments to identify the food substances present in specimens O and M. Your experimental report should be tabulated as shown.

Identification of Food Substances in Specimens O and M

Food Tested	**Procedure**	**Observation**	**Inference**
Starch (Specimen O)	Add a few drops of iodine solution to specimen O solution	Blue black colour observed	Starch is present
Reducing sugar (Specimen M)	Add Benedict's solution to specimen M solution and heat	Brick red precipitate formed	Reducing sugar is present
Protein (Specimen M)	Add sodium hydroxide then copper sulfate solution	Violet or purple colour formed	Protein is present

(c) State the function(s) of each food substance identified in 1(b) to the human body.

Starch provides energy to the body. It is broken down into glucose, which is used during respiration to release energy for body activities.

Reducing sugars supply quick energy to the body. They are easily absorbed into the bloodstream and used immediately by cells.

Proteins are essential for growth and repair of body tissues. They are also used in the formation of enzymes, hormones, and antibodies.

(d) Apart from the specimen provided give another one example of natural source from which the food substances identified in 1(b) could be obtained.

Starch can be obtained from natural sources such as maize or cassava.

Reducing sugars can be obtained from fruits such as ripe bananas or sugarcane.

Proteins can be obtained from beans, eggs, or meat.

2. You have been provided with specimens E F and G obtained from different habitat. Carefully study them and answer the following questions.

(a) Identify the Phylum or Division and the Kingdom in which each specimen E F and G belongs.

Specimen E belongs to Phylum Arthropoda and Kingdom Animalia.

Specimen F belongs to Division Angiospermae and Kingdom Plantae.

Specimen G belongs to Division Fungi and Kingdom Fungi.

(b) For each of specimens E F and G state three observable features which are characteristic of its respective Phylum or Division.

Specimen E has a segmented body, jointed appendages, and a hard exoskeleton made of chitin.

Specimen F has true roots, stems, and leaves. It also produces flowers and seeds enclosed in fruits.

Specimen G lacks chlorophyll, has thread like hyphae, and feeds by saprophytic nutrition.

(c) Name the habitat of each of specimens E F and G.

Specimen E lives in terrestrial habitat.

Specimen F lives in terrestrial habitat.

Specimen G lives in moist or decaying organic matter.

(d) For each of specimens E F and G state whether it is a producer consumer or decomposer in its natural habitat. Give reason for each case.

Specimen E is a consumer because it feeds on other organisms for energy.

Specimen F is a producer because it manufactures its own food through photosynthesis.

Specimen G is a decomposer because it feeds on dead organic matter and breaks it down into simpler substances.

(e) State two advantages and one disadvantage of specimen E to other living organisms.

One advantage of specimen E is that it serves as a source of food to other organisms in the food chain.

Another advantage is that it helps in pollination or soil aeration depending on the species.

One disadvantage is that some arthropods act as pests or disease vectors, causing harm to plants and animals.

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CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/2B **BIOLOGY 2B**

(ACTUAL PRACTICAL B)

(For Both School and Private Candidates)

Duration: 2:30 Hours **SOLUTIONS** **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writing should be in **blue** or **black** ink, except drawing which must be in pencil.
4. Communication devices and any unauthorised materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)

1. (a) You have been provided with solutions P and Q each containing a mixture of food substances. Carry out the experiment to identify the food substances present in the solutions provided. Your experimental report should be tabulated as shown in Table 1.

Table 1: Identification of Food Substances in the Solutions P and Q

Food Tested	**Procedure**	**Observation**	**Inference**
Starch (P)	Add a few drops of iodine solution to solution P	Blue black colour observed	Starch present
Reducing sugar (P)	Add Benedict's solution to solution P and heat	Brick red precipitate formed	Reducing sugar present
Protein (Q)	Add sodium hydroxide then copper sulfate to solution Q	Violet or purple colour formed	Protein present
Reducing sugar (Q)	Add Benedict's solution to solution Q and heat	Green to brick red precipitate observed	Reducing sugar present

(b) Specimen N contains lipid among others. Prepare the specimen for the lipid test experiment using a filter paper then outline the procedure used to prepare it.

A small amount of specimen N is crushed to release its contents. The crushed specimen is gently rubbed onto a clean dry filter paper to allow any lipid present to soak into the paper. The filter paper is then left to dry.

(c) Use the filter paper provided to identify the presence of lipid in the specimen N. Present your findings as shown in Table 2.

Table 2: Identification of Lipid in Specimen N

Food Tested	Procedure	Observation	Inference
Lipid	Rub specimen N on filter paper and allow it to dry	Translucent greasy spot observed	Lipid present

(d) State the function(s) of each food substance identified in the solutions and specimen N to the human body.

Starch provides energy to the body. It is broken down into glucose which is used during respiration to supply energy for growth, movement, and body processes.

Reducing sugars supply immediate energy to the body. They are rapidly absorbed into the bloodstream and used by cells for quick energy needs.

Proteins are essential for growth and repair of body tissues. They are also used to make enzymes, hormones, and antibodies.

Lipids act as a concentrated source of energy. They also provide insulation to reduce heat loss and protect vital organs.

(e) Why boiling was important in some procedures of the experiments conducted?

Boiling was important to speed up chemical reactions and allow reagents such as Benedict's solution to react effectively with food substances. It also helps to make some substances more available for testing.

2. You have been provided with specimens I J K and L. Carefully study them and answer the following questions.

(a) State two distinctive features of each specimen I J and K at Class level.

Specimen I has a segmented body and jointed appendages. It also has a hard exoskeleton made of chitin.

Specimen J has feathers covering its body and forelimbs modified into wings.

Specimen K has moist skin and lives both on land and in water.

(b) Specimens I and J belong to the same major group. State two characteristics they have in common.

Both specimens have a backbone, showing they are vertebrates. They also have an internal skeleton which supports the body.

(c) How do the specimens I and L adopted to survive in their natural habitats? Give one point for each.

Specimen I survives by having jointed limbs that allow efficient movement in search of food and shelter.

Specimen L survives by having broad leaves which increase surface area for photosynthesis.

(d) State two ways the farmers benefit from the availability of the specimen L in their farms.

Specimen L improves soil fertility by adding organic matter when its leaves decay.

It also provides shade or acts as a windbreak, protecting crops from harsh environmental conditions.

(e) Outline three advantages and one disadvantage of the specimen I in nature.

Specimen I serves as a source of food for other organisms, helping to maintain food chains.

It helps in pollination or decomposition depending on the species, contributing to ecosystem balance.

Specimen I assists in soil aeration through movement in the soil.

One disadvantage of specimen I is that some species act as pests or disease vectors, causing harm to crops and humans.

THE UNITED REPUBLIC OF TANZANIA

NATIONAL EXAMINATIONS COUNCIL OF TANZANIA

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

033/2C **BIOLOGY 2C**

(ACTUAL PRACTICAL C)

(For Both School and Private Candidates)

Duration: 2:30 Hours **SOLUTIONS** **Year: 2025**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks.
3. All writing should be in **blue** or **black** ink, except drawing which must be in pencil.
4. Communication devices and any unauthorised materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)

1. (a) You have been provided with solutions A and B which contain the mixture of different food substances. Carry out an experiment to identify the food substances present in each solution. Your experimental report should be tabulated as shown in Table 1.

Table 1: Identification of Food Substances in Solution A and B

Food tested	**Procedure**	**Observation**	**Inference**
Starch (A)	Add a few drops of iodine solution to solution A	Blue black colour observed	Starch present
Reducing sugar (A)	Add Benedict's solution to solution A and heat	Brick red precipitate formed	Reducing sugar present
Reducing sugar (B)	Add Benedict's solution to solution B and heat	Green to brick red precipitate observed	Reducing sugar present
Protein (B)	Add sodium hydroxide then copper sulfate to solution B	Violet or purple colour observed	Protein present

(b) Specimen C contains protein among others. Prepare the solution from it for Biuret test experiment then state the procedure followed to prepare it.

A small amount of specimen C is crushed thoroughly using a mortar and pestle. Distilled water is added to the crushed specimen and the mixture is stirred well. The mixture is then filtered to obtain a clear solution which is used for the Biuret test.

(c) Carry out the Biuret test to identify food substances present in specimen C. Your experimental report should be tabulated as shown in Table 2.

Table 2: Identification of Biuret Test for Food in Specimen C

Food Tested	**Procedure**	**Observation**	**Inference**
Protein	Add sodium hydroxide to the solution of specimen C, then add copper sulfate dropwise	Violet or purple colour formed	Protein present

(d) State the function(s) of each food substances identified in solutions A B and specimen C to the human body.

Starch supplies energy to the body. It is broken down into glucose which is used during respiration to provide energy for body activities.

Reducing sugars provide quick and readily available energy. They are rapidly absorbed into the bloodstream and used by body cells.

Proteins are used for growth and repair of body tissues. They are also required for the formation of enzymes, hormones, and antibodies.

(e) “Sodium hydroxide” and “dilute hydrochloric acid” were important in some procedure of the experiments. Justify this statement.

Sodium hydroxide is important in the Biuret test because it provides an alkaline medium that allows proteins to react with copper sulfate to form a violet colour. Dilute hydrochloric acid is used to break down complex food substances into simpler forms, making them easier to test and detect during food experiments.

2. You have been provided with specimens W X Y and Z obtained from different habitats. Carefully study them then answer the following questions.

   (a) Identify the Kingdom Phylum or Division and Class in which each specimen W X Y and Z belongs.

Specimen W belongs to Kingdom Animalia, Phylum Arthropoda, Class Insecta.

Specimen X belongs to Kingdom Animalia, Phylum Chordata, Class Pisces.

Specimen Y belongs to Kingdom Animalia, Phylum Chordata, Class Amphibia.

Specimen Z belongs to Kingdom Plantae, Division Angiospermophyta, Class Dicotyledonae.

   (b) For each specimen W X Y and Z state three visible features which are the characteristics of the respective Class it belongs.

Specimen W has a segmented body divided into head, thorax, and abdomen. It has three pairs of jointed legs and a pair of antennae.

Specimen X has fins for swimming, a streamlined body, and scales covering the body.

Specimen Y has moist smooth skin, strong hind limbs for jumping, and lives both on land and in water.

Specimen Z has broad leaves with reticulate venation, a tap root system, and flowers with parts in multiples of four or five.

(c) Specimens W and X belongs to the same Kingdom and Phylum. Which characteristics they have in common to represent those groups?

Both specimens have an internal body organization and respond to stimuli, showing characteristics of Kingdom Animalia. They also have bilateral symmetry and well developed organ systems.

(d) For each specimen W and Y state two advantages and one disadvantage they have for other living organisms.

Specimen W helps in pollination of flowering plants. It also serves as food for other animals. One disadvantage is that it may act as a pest or disease vector.

Specimen Y helps control insect populations. It also serves as food for higher predators. One disadvantage is that it may transmit parasites or cause imbalance when introduced into new habitats.